Control Statements Loop

```
#range
range(0,10)
```

```
range(0, 10)
```

```
a=list(range(10))
print(a)
```

```
[0, 1, 2, 3, 4, 5, 6, 7, 8, 9]
```

```
a=list(range(0,10))
print(a)
a=list(range(1,10,2))
print(a)
```

```
[0, 1, 2, 3, 4, 5, 6, 7, 8, 9]
[1, 3, 5, 7, 9]
```

```
print(range(0,20))
print(list(range(0,20)))
print(list(range(0,20,3)))
```

```
range(0, 20)
[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19]
[0, 3, 6, 9, 12, 15, 18]
```

for loop

```
#syntax : for Val in sequence :
numbers = [6,5,3,8,4,2,5,4,11]
sum=0
for val in numbers:
    sum = sum + val
print("The sum is ",sum)
```

```
The sum is  48
```

for loop with else

```
list_of_digit = [0,1,2,3,4,5,6]
input_digit = int(input("Enter a digit : "))
for i in list_of_digit :
    if (i == input_digit ):
        print("Digit is in the list")
```

```
        break
else :
        print("Digit not found")
```

```
Enter a digit : 55
Digit not found
```

```
list_of_digit = [0,1,2,3,4,5,6]
input_digit = int(input("Enter a digit : "))
for i in list_of_digit :
    if (i == input_digit ):
        print("Digit is in the list")
else :
        print("Digit not found")
```

```
Enter a digit : 2
Digit is in the list
Digit not found
```

```
sum = 0
for i in range(11):
    sum += i
print(sum)
```

```
55
```

```
n1 = 5
n2 = 5
for i in range(n1):
    for j in range(n2):
        print("*",end=" ")
    print()
```

```
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
```

```
n1 = 5
n2 = 5
for i in range(n1):
    for j in range(n2):
        print("* ",end=" ")
    print()
```

```
*  *  *  *  *
*  *  *  *  *
*  *  *  *  *
*  *  *  *  *
*  *  *  *  *
```

```python
n1 = 5
n2 = 5
for i in range(n1):
    for j in range(i,n1):
        print("*",end=" ")
    print()
```

```
* * * * *
* * * *
* * *
* *
*
```

```python
n1 = 5
n2 = 5
for i in range(n1):
    for j in range(i):
        print("*",end=" ")
    print()
```

```

*
* *
* * *
* * * *
```

```python
n1 = 5
n2 = 5
for i in range(n1):
    for j in range(i+1):
        print("*",end=" ")
    print()
```

```
*
* *
* * *
* * * *
* * * * *
```

```python
n1 = 5
n2 = 5
for i in range(n1):
    for j in range(i+1):
        print("*",end="")
    print()
```

```
*
**
***
****
*****
```

```
n1 = 5
n2 = 5
for i in range(n1):
    for j in range(i,n1):
        print(" ",end="")
    for k in range(i+1):
        print("*",end="")
    print()
```

```
     *
    **
   ***
  ****
 *****
```

```
n1 = 5
n2 = 5
for i in range(n1):
    for j in range(i+1):
        print("*",end="")
    print()

for i in range(n1):
    for j in range(i,n1):
        print("*",end="")
    print()
for i in range(n1):
    for j in range(i,n1):
        print(" ",end="")
    for k in range(i+1):
        print("*",end="")
    print()
```

```
*
**
***
****
*****
*****
****
***
**
*
     *
    **
   ***
  ****
 *****
```

While Loop

```
n = int(input("Enter n : "))
i=1
sum = 0
while i<=n:
    sum += i
    i +=1
print(sum)
```

```
Enter n : 11
66
```

```
counter = 0
while counter < 3:
    print("Inside Loop")
    counter += 1
else :
    print("Inside Else")
```

```
Inside Loop
Inside Loop
Inside Loop
Inside Else
```

```
n = int(input("Enter a number : "))
i = 2
count = 0

while (i < n):
    if (n % i == 0):
        count += 1
        break  # Move break inside the if-condition to stop
immediately if a divisor is found
    i += 1     # Increment i at the while-loop level, not inside the
if-statement

if (count == 0 and n > 1):  # A prime number has 0 divisors between 2
and n-1, and must be > 1
    print("Prime")
else:
    print("Not prime")
```

```
Enter a number : 7
Prime
```

```
for i in range (1,51):
    if(i%2!=0):
        continue
    if(i%2==0):
        print(i,end=" ")
```

```
2 4 6 8 10 12 14 16 18 20 22 24 26 28 30 32 34 36 38 40 42 44 46 48 50
```

```
c=0
while c<10:
    if c==3:
        print("using continue stateemnt : skipping iteration when
count is : ",c)
        c=c+1
        continue
    if c==6:
        print("using pass statement : count is : ",c)
        c=c+1
        pass
    if c==8:
        print("using break statement : count is : ",c)
        c=c+1
        break
    print("Current count is : ",c)
    c=c+1
```

```
Current count is :  0
Current count is :  1
Current count is :  2
using continue stateemnt : skipping iteration when count is :  3
Current count is :  4
Current count is :  5
using pass statement : count is :  6
Current count is :  7
using break statement : count is :  8
```

```
for i in range(1,21,2):
    print(i)
```

```
1
3
5
7
9
11
13
15
17
19
```

```
for i in range(6):
    print(i)
```

```
0
1
2
3
```

```
4
5

# 1 to 10
for i in range(1,11):
    print(i,end=" ")

1 2 3 4 5 6 7 8 9 10

# 10 to 1
for i in range(10,0,-1):
    print(i,end=" ")

10 9 8 7 6 5 4 3 2 1

# table of
n = int(input("Enter n: "))
for i in range(1,11):
    print(f"{n} X {i} = {n*i}")

Enter n: 5
5 X 1 = 5
5 X 2 = 10
5 X 3 = 15
5 X 4 = 20
5 X 5 = 25
5 X 6 = 30
5 X 7 = 35
5 X 8 = 40
5 X 9 = 45
5 X 10 = 50

# factorial of a num
n = int(input("Enter n : "))
fact = 1
for i in range(1,n+1):
    fact = fact * i
print(fact)

Enter n : 5
120

# sum of digits
n = int(input("Enter n : "))
sum =0
while(n!=0) :
    sum = sum + int(n%10)
    n=int(n/10)
print(sum)
```

```
Enter n : 123
6

# Reverse of a number
n = int(input("Enter n : "))
rn = 0

while n != 0:
    rn = (rn * 10) + (n % 10)
    n = n // 10

print(rn)

Enter n : 445
544

#palindrome
# Reverse of a number
n = int(input("Enter n : "))
n1 = n
rn = 0

while n != 0:
    rn = (rn * 10) + (n % 10)
    n = n // 10  # Integer division to reduce n

if (rn == n1):
    print("palindrome")
else :
    print("not palindrome")

Enter n : 252
palindrome

# fibonacci
a = 0
b = 1
n = int(input("enter no of elements in series u want : "))
for i in range (n):
    print(f"{a}",end=" ")
    a,b = b,a+b

enter no of elements in series u want : 6
0 1 1 2 3 5

#traverse string

s = input("Enter a string")

for i in range(len(s)):
    print(f"{i} -> {s[i]}")
```

```
Enter a stringraphael
0 -> r
1 -> a
2 -> p
3 -> h
4 -> a
5 -> e
6 -> l
```

```
#pattern 1
for i in range(1,6):
    for k in range(6-i):
        print(" ",end=" ")
    for j in range(i):
        print("*",end=" ")
    print()
```

```
          *
        * *
      * * *
    * * * *
  * * * * *
```

```
#pattern 2
for i in range(5):
    for j in range(1,6):
        print(j,end="")
    print()
```

```
12345
12345
12345
12345
12345
```

```
#pattern 3
for i in range(1,6):
    for j in range(1,i+1):
        print(j,end="")
    print()
```

```
1
12
123
1234
12345
```

```
#patttern 4
k=65
for i in range(1,6):
    for j in range(1,i+1):
```

```
        print(chr(k),end="")
        k+=1
    print()
```

```
A
BC
DEF
GHIJ
KLMNO
```

```
#patttern 5
k=65
for i in range(1,6):
    for j in range(1,i+1):
        print(chr(k),end="")
    k+=1
    print()
```

```
A
BB
CCC
DDDD
EEEEE
```